

Rajalakshmi Engineering College

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NeoColab_REC_CS23231_DATA STRUCTURES

REC_DS using C_Week 1_COD_Question 2

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Arun is learning about data structures and algorithms. He needs your help in solving a specific problem related to a singly linked list.

Your task is to implement a program to delete a node at a given position. If the position is valid, the program should perform the deletion; otherwise, it should display an appropriate message.

Input Format

The first line of input consists of an integer N, representing the number of elements in the linked list.

The second line consists of N space-separated elements of the linked list.

The third line consists of an integer x, representing the position to delete.

Position starts from 1.

Output Format

The output prints space-separated integers, representing the updated linked list after deleting the element at the given position.

If the position is not valid, print "Invalid position. Deletion not possible."

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5

8 2 3 1 7

2

Output: 8 3 1 7

Answer

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
void insert(int);
```

```
void display_List();
```

```
void deleteNode(int);
```

```
struct node {
```

```
    int data;
```

```
    struct node* next;
```

```
} *head = NULL, *tail = NULL;
```

```
void insert(int element){
```

```
    struct node* new_node=(struct node*)malloc(sizeof(struct node));
```

```
    new_node->data=element;
```

```
    new_node->next=NULL;
```

```
    if(head==NULL){
```

```
        head=new_node;
```

```
        tail=new_node;
```

```
    }
```

```
    else{
```

```
        tail->next=new_node;
```

```

        tail=new_node;
    }
}

void deleteNode(int pos){
    if(head==NULL||pos<=0){
        printf("Invalid position. Deletion not possible.\n");
        return;
    }
    struct node* temp=head;
    if(pos==1){
        head=head->next;
        free(temp);
        if(head==NULL){
            tail=NULL;
        }
    }
    else{
        struct node* prev=NULL;
        for(int i=1;temp!=NULL&& i<pos;i++){
            prev=temp;
            temp=temp->next;
        }
        if(temp==NULL){
            printf("Invalid position. Deletion not possible.\n");
            return;
        }
        prev->next=temp->next;
        if(temp==tail){
            tail=prev;
        }
        free(temp);
    }
    display_List();
}

```

```

void display_List(){
    struct node* temp=head;
    if(temp==NULL){
        printf("\n");
        return;
    }
}

```

```
while(temp!=NULL){
    printf("%d ",temp->data);
    temp=temp->next;
}
printf("\n");
}

int main() {
    int num_elements, element, pos_to_delete;

    scanf("%d", &num_elements);

    for (int i = 0; i < num_elements; i++) {
        scanf("%d", &element);
        insert(element);
    }

    scanf("%d", &pos_to_delete);

    deleteNode(pos_to_delete);

    return 0;
}
```

Status : Correct

Marks : 10/10